Engine Model(s): PT6A

72-40-01

COMBUSTION CHAMBER ASSEMBLY - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-151	Wire, Filler
PWC11-014	Alcohol, Isopropyl

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

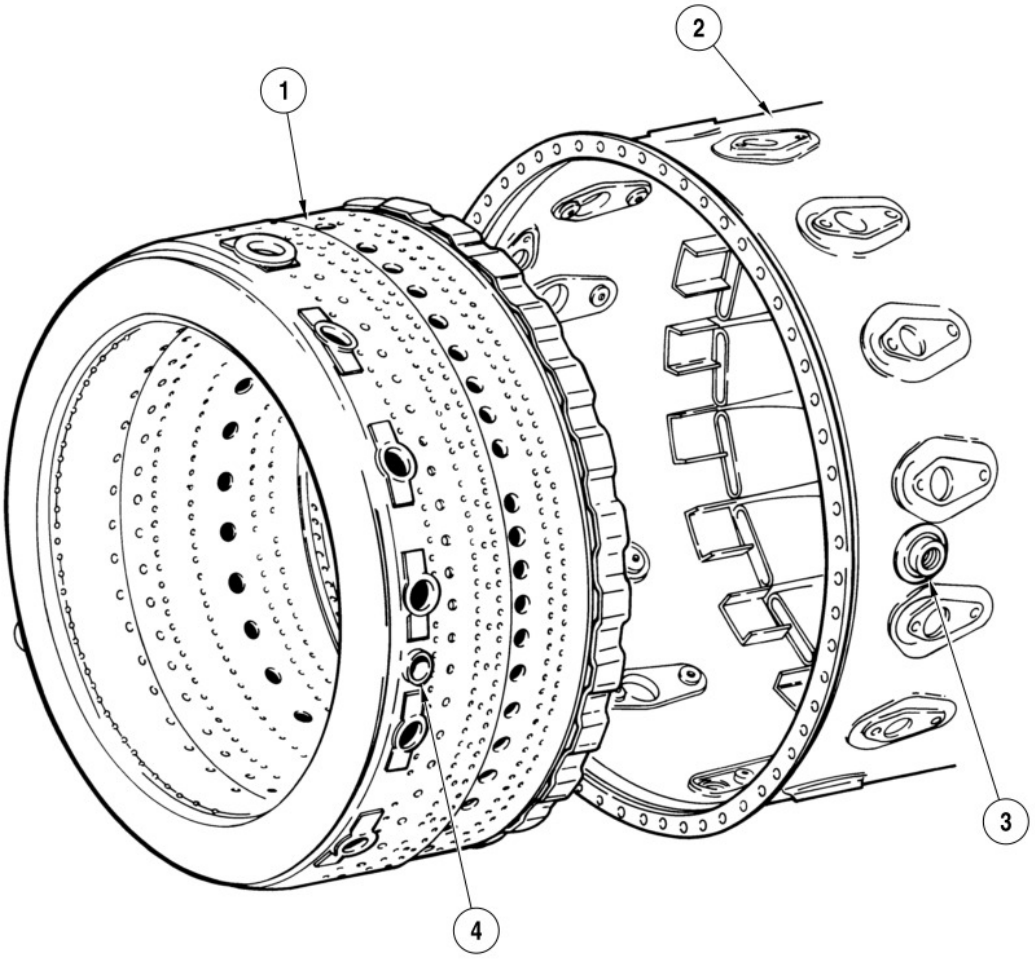
Not Applicable

5. Removal/Installation

A. Removal of Combustion Chamber Liner (Ref. Fig. 201)

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Figure 201 REMOVAL/INSTALLATION of Combustion Chamber Liner



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1. Combustion Chamber Liner
2. Gas Generator Assembly
3. Gas Generator Case Glow Plug/Spark Igniter Bosses
4. Combustion Chamber Liner Glow Plug/Spark Igniter Bosses

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(1) Remove the power section (Ref. 72-00-00, SERVICING).

(2) Remove the fuel manifold adapter assemblies (Ref. 73-10-05, REMOVAL/INSTALLATION).

(3) Remove the ignition glow plugs (Ref. 74-20-02) or spark igniters (Ref. 74-20-04, REMOVAL/INSTALLATION).

CAUTION: DO NOT ATTEMPT TO PRY THE LINER LOOSE WITH SHARP EDGED OR POINTED TOOLS.

(4) Withdraw the combustion chamber liner (1) from the gas generator assembly (2).

NOTE: If difficulty is experienced in withdrawing the liner from the gas generator assembly, strike the inside wall of the liner with the palm of your hand to dislodge the liner from the small and large exit duct joints.

B. Installation of Combustion Chamber Liner (Ref. Fig. 201)

(1) Install the combustion chamber liner (1) in the gas generator assembly (2), aligning the glow plug/spark igniter bosses (4) in the outer wall of the liner with the corresponding bosses (3) on the gas generator assembly.

(2) Make sure that the rear flanges Pre-SB1434 or rear sliding joint flanges Post-SB1434 (Ref. Fig. 208) of the liner inner and outer walls engage with the corresponding flanges on the small and large exit ducts.

(3) Install the ignition glow plugs (Ref. 74-20-02) or spark igniters (Ref. 74-20-04, REMOVAL/INSTALLATION).

(4) Install the fuel manifold adapter assemblies (Ref. 73-10-05, REMOVAL/INSTALLATION).

(5) Install the power section on the gas generator assembly (Ref. 72-00-00, SERVICING).

6. Inspection/Check

A. General

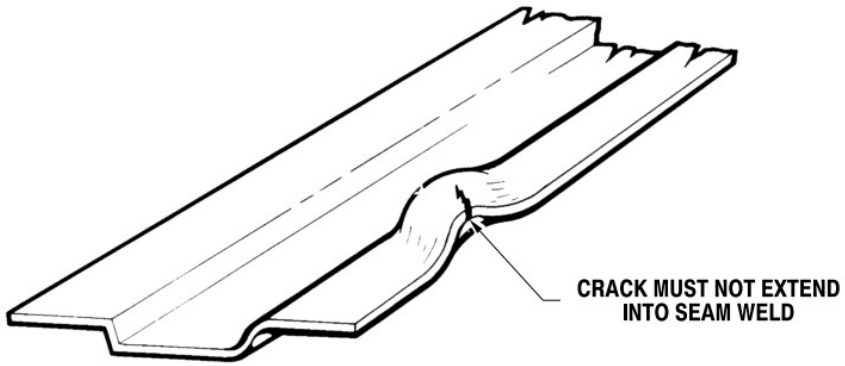
(1) Examine the combustion chamber liners for cracks, buckling and burning. Cracks in the liner surface are usually of a stress relieving nature and are usually not serious, since the rate of growth decreases as the crack lengthens. Thermal stresses, in effect, relieve original stress conditions. It is normal to observe a given type of deterioration repeated from liner to liner in the same engine, or engines of the same model. Typical liner distress consists of buckling and cracking of the cooling rings and cracking at the inner wall of the exit duct section of the outer liner.

B. Acceptable Conditions, Repair Unnecessary

(1) Localized buckling and/or burning of the cooling rings, except the louvered type, accompanied by cracking is acceptable provided the cracks do not extend into the seam weld (Ref. Fig. 202).

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Figure 202 Acceptable Buckling and Cracking in Cooling Rings (Non-Louvered Type)



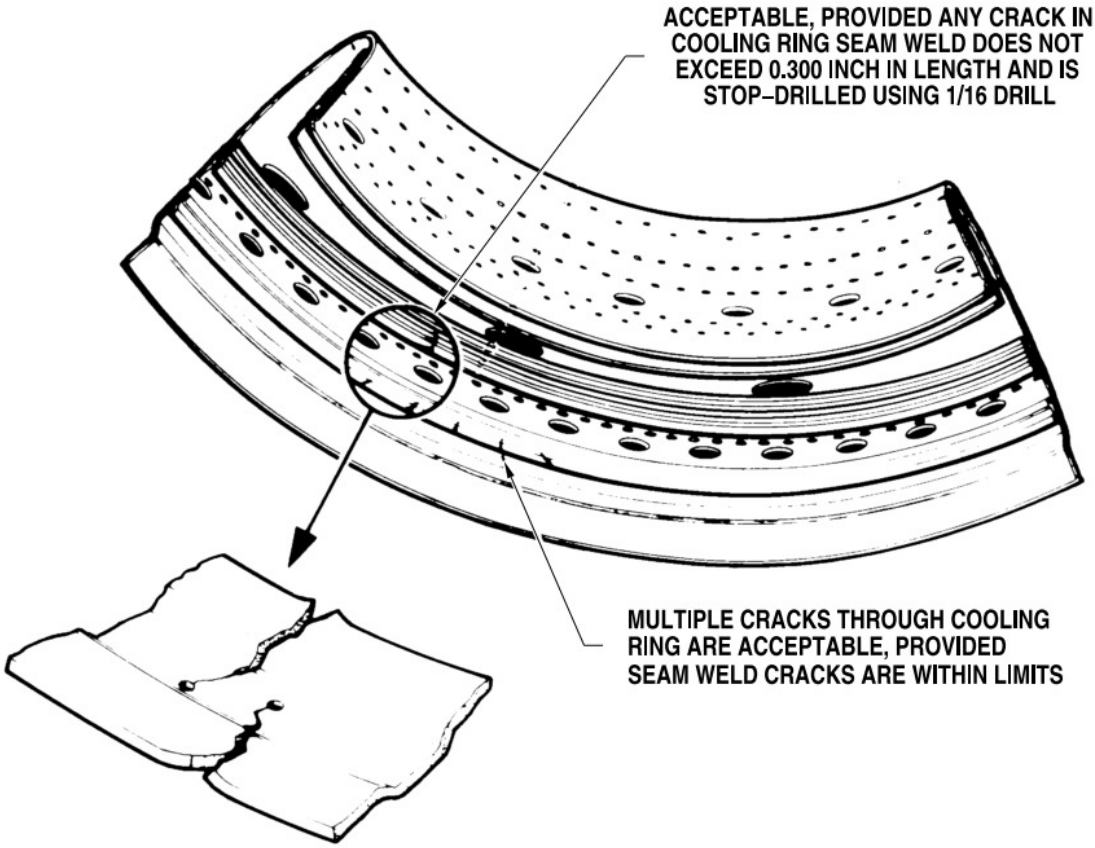
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(2) Cracks in the louvered type cooling ring (Ref. Fig. 203) are acceptable provided that circumferential cracks in the seam weld (Ref. Fig. 204) not exceeding 0.300 inch in length are stop drilled using a 1/16 (0.0625) inch drill.

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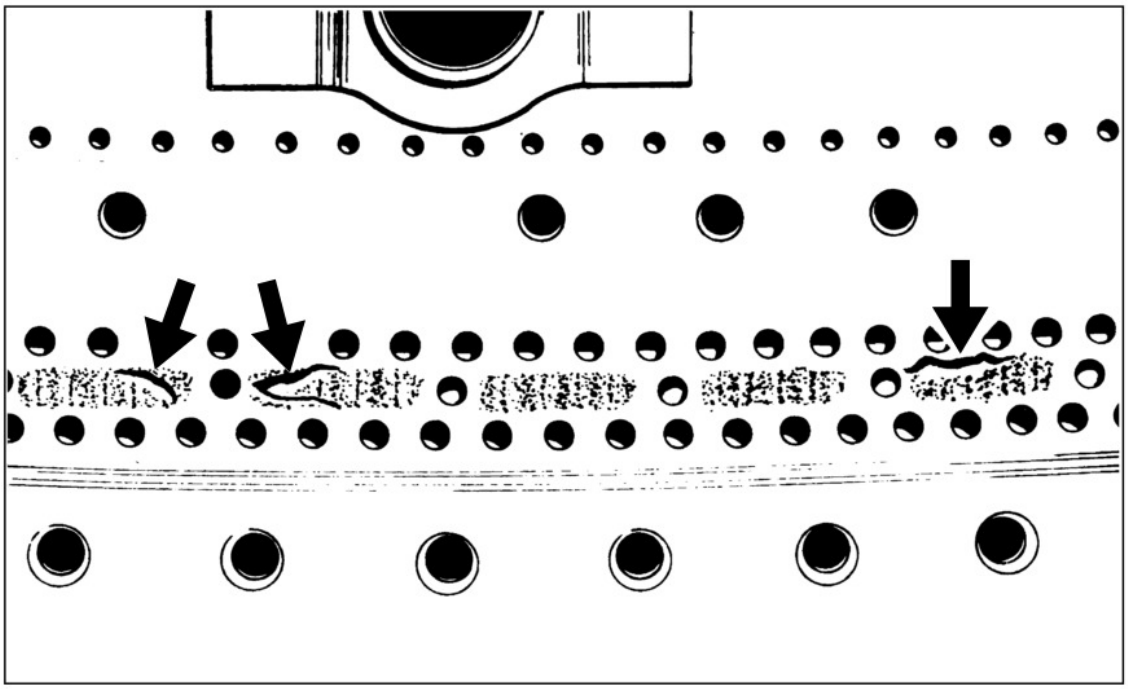
Figure 203 Acceptable Cracks in Louvered Type Cooling Ring



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Figure 204 Acceptable Cracks in Louvered Cooling Ring Seam Weld



CRACKS OF THIS NATURE ARE ACCEPTABLE, PROVIDED THEY DO NOT EXCEED 0.300 INCH IN LENGTH AND ARE STOP-DRILLED USING 1/16 DRILL.

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(3) Straight line cracks between two adjacent cooling holes are acceptable.

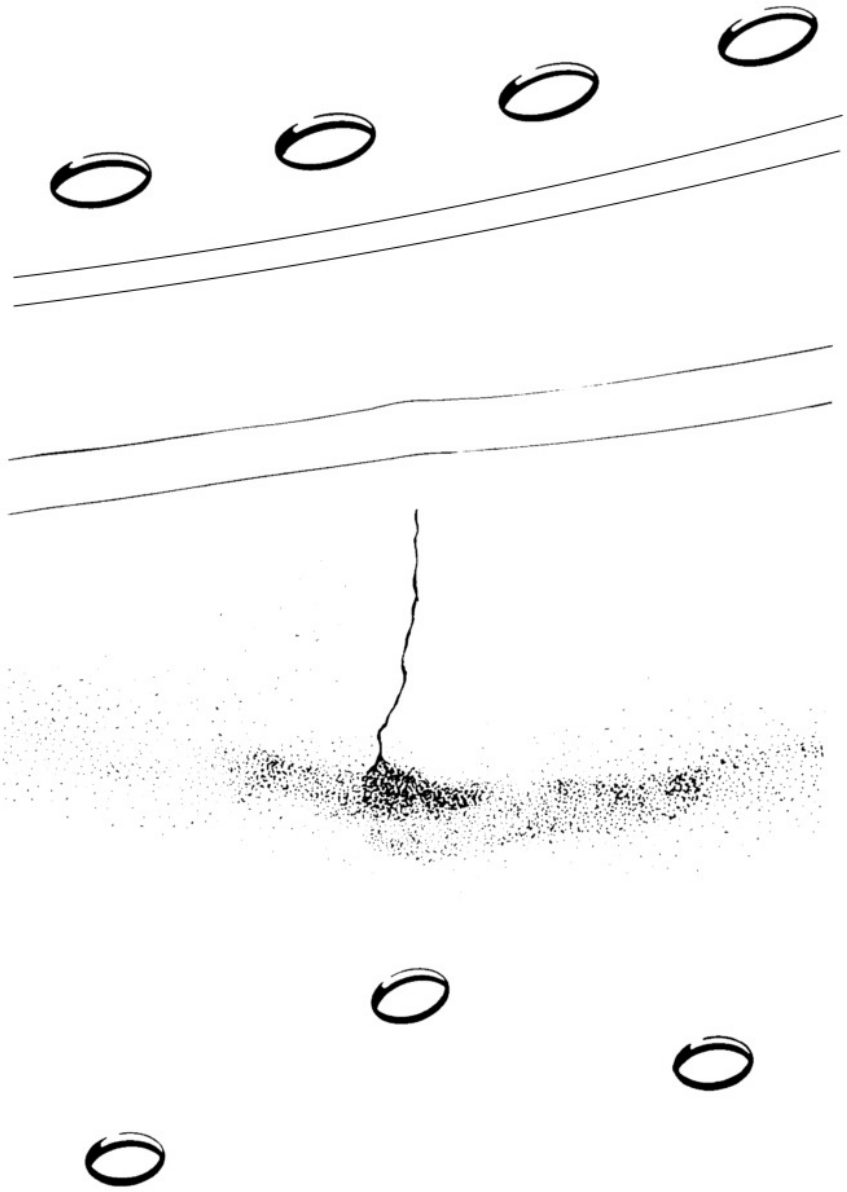
CAUTION: CRACKS RUNNING IN THE SAME DIRECTION, AND WHICH COULD ULTIMATELY JOIN, ARE NOT ACCEPTABLE UNLESS THEY ARE SEPARATED BY AT LEAST THREE INCHES OF PARENT METAL.

(4) A maximum of seven cracks, each crack not exceeding one inch long, is acceptable in the inner wall of the domed end of the inner liner (Ref. Fig. 205).

(5) Inspect convolutions (Pre-SB1434) of support strip on outer liner for wear. Wear not exceeding 50 percent of material thickness is acceptable. If convolutions worn in excess of 50 percent, ship the liner to an approved overhaul facility for repair.

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Figure 205 Acceptable Cracks in Inner Wall of Liner



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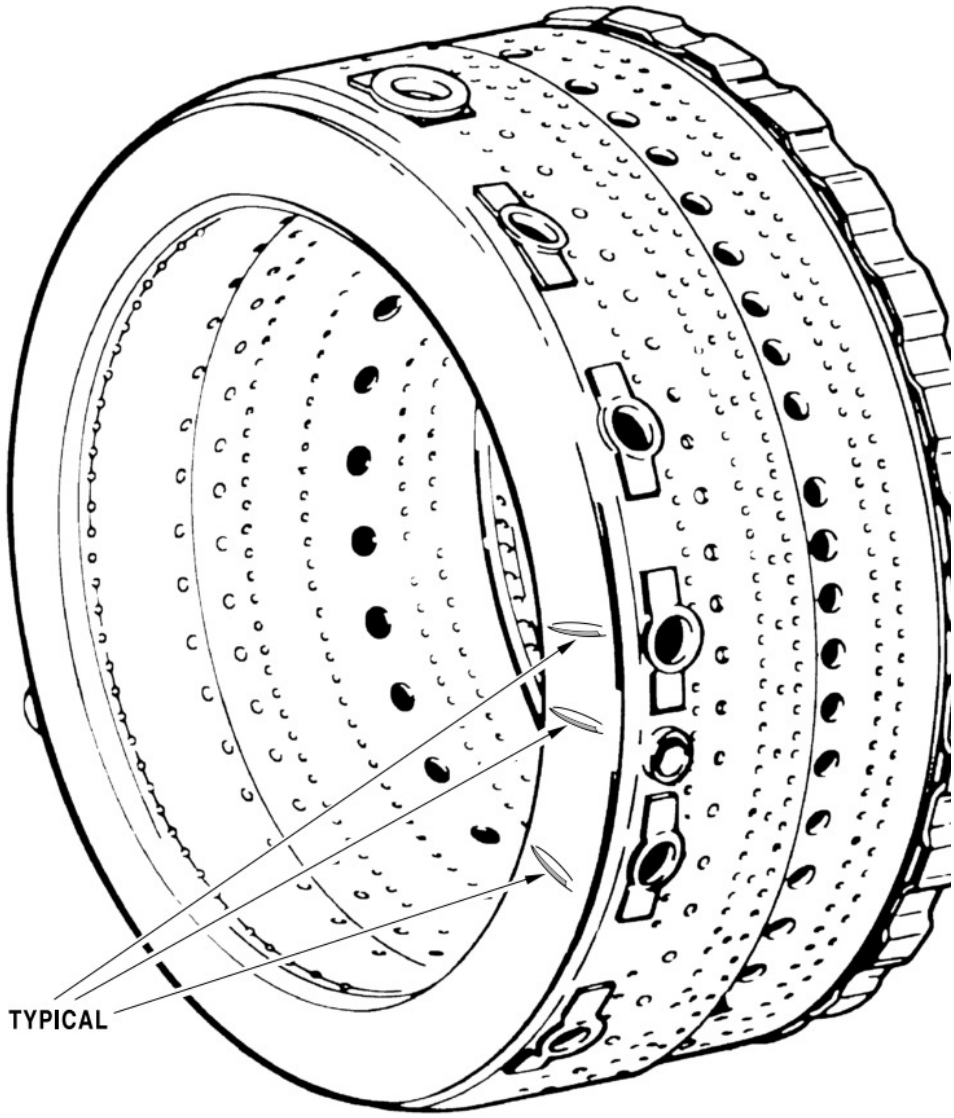
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(6) Localized areas that have been heated to an extent as to cause buckling of the liner are considered acceptable provided the buckling is shallow, and it is not associated with burning, which would reduce the wall thickness and weaken the structure (Ref. Fig. 207).

(7) The dome bulging is often associated with cooling ring buckling and reduction of cooling ring gap, and repair of the gap is necessary (Ref. Fig. 206).

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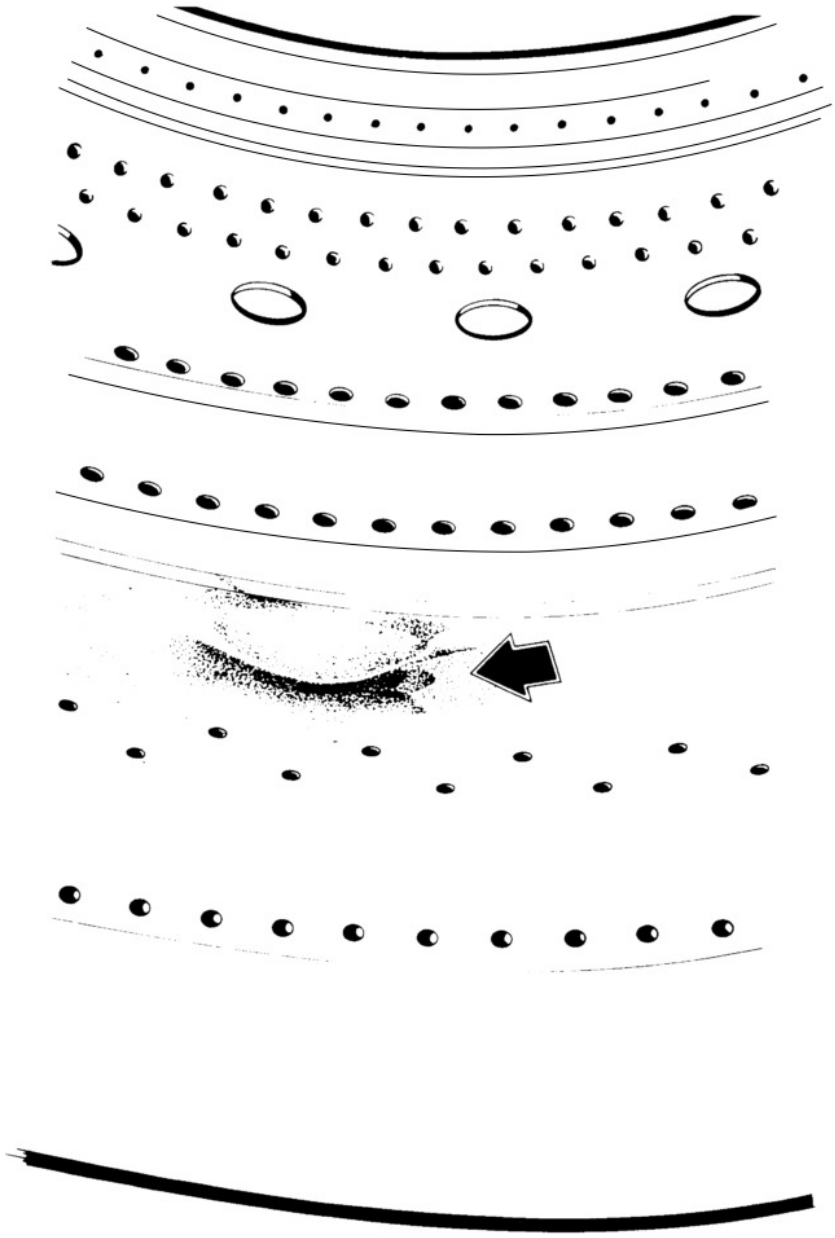
Figure 206 Acceptable Bulging in Liner Dome



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Figure 207 Acceptable Buckling in Liner



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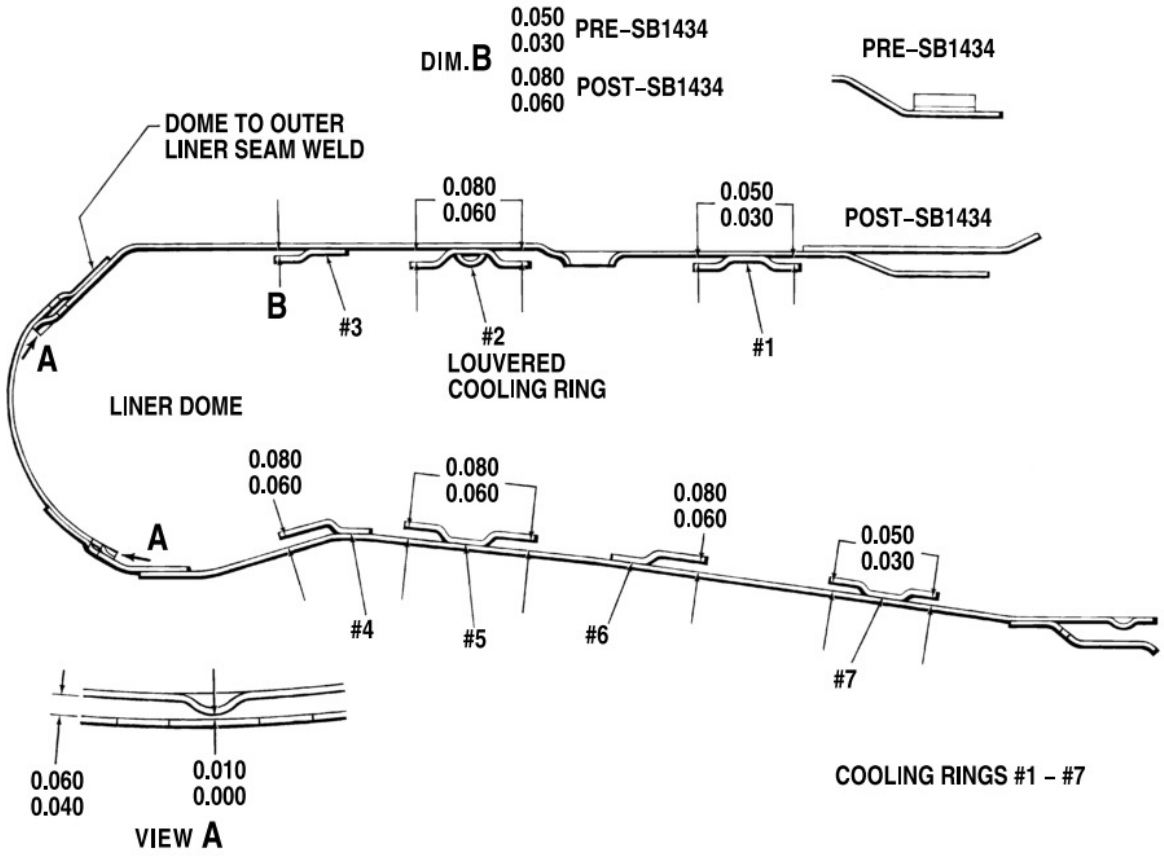
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C. Acceptable Conditions, Repair Necessary

(1) Buckled cooling rings and liner dome, where the gap has been eliminated, must be reworked. Use a suitable bar to restore a uniform gap to dimensions shown (Ref. Fig. 208).

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Figure 208 Restoration of Gaps in Liner Dome and Cooling Rings



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(2) Cracks not exceeding two inches in length must be repaired (Ref. Para. 7.).

(3) Circumferential cracks next to seam welds must be repaired (Ref. Para. 7.).

NOTE: Crack openings in excess of 0.030 inch are not repairable.

(4) When cracks progress from a free edge so that their meeting is imminent, and could subsequently allow a piece of liner material to break loose, they must be repaired by welding (Ref. Para. 7.).

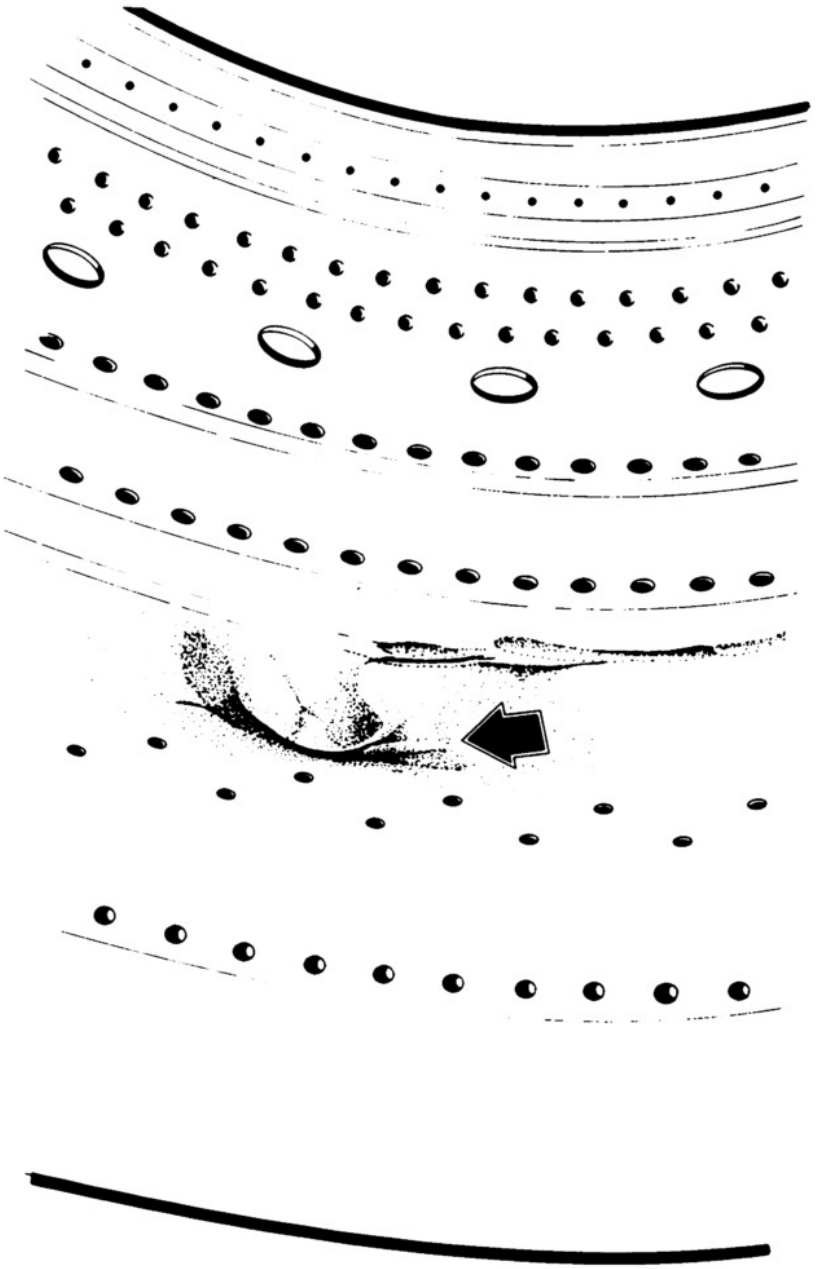
D. Unacceptable Conditions, No Repair Permitted

NOTE: Rejected combustion chamber inner and outer liners removed during maintenance activities should be returned to an overhaul facility for possible restoration to a usable condition. It is recommended that "run" inner and outer liners be kept together in pairs, if possible.

(1) Severe buckling causing kinked metal (Ref. Fig. 209).

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Figure 209 Unacceptable Buckling in Liner



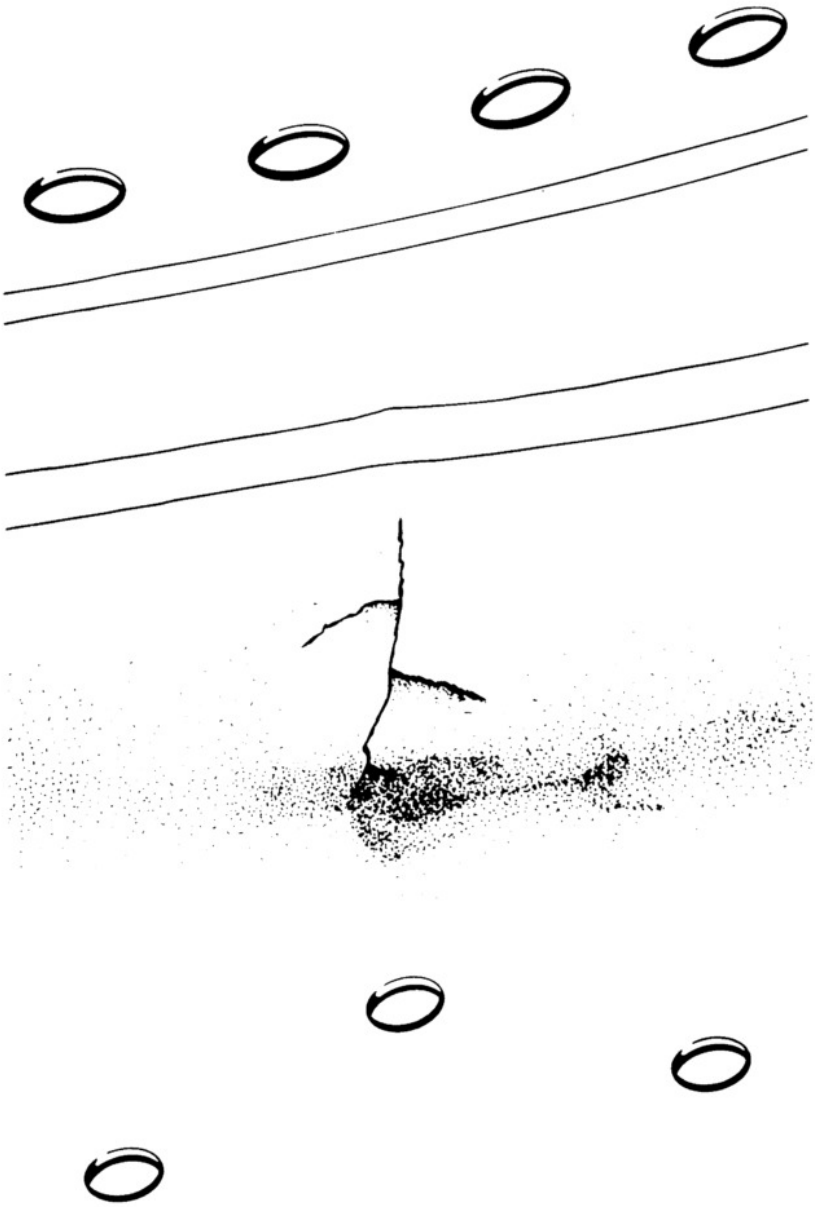
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(2) Multiple cracks are not repairable (Ref. Fig. 210).

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Figure 210 Unacceptable Multiple Cracks in Liner



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(3) Cracks exceeding two inches in length are not repairable.

(4) Circumferential cracks which are opened more than 0.030 inch and are next to seam welds are not repairable.

E. Combustion Chamber Large Exit Duct

(1) Inspect large (outer) exit duct for cracks, burning, distortion or fretting wear on the inner diameter which contacts the combustion chamber liner. Any amount of coating loss in contact area is acceptable. Coating loss in other areas acceptable provided parent metal erosion (if present) is not in excess of 0.010 in. deep. No repairs authorized at maintenance level.

7. Approved Repairs

A. Repair of Cracks

CAUTION: HOLES THAT BECOME PLUGGED AS A RESULT OF WELDING MUST BE REDRILLED TO THE ORIGINAL SIZE AND THEN DEBURRED. CHECK THE PATTERN AND SIZE OF THE ADJACENT HOLES USING A SUITABLE DRILL GAGE. LOOSE WELD MATERIAL MUST BE REMOVED.

(1) Stop drill the ends of a crack with a $^{1}/_{16}$ (0.0625) inch drill. Vee out a crack using a suitable abrasive wheel.

(2) Thoroughly wire brush the area and swab with alcohol (PWC11-014).

(3) Weld the prepared area using tungsten inert gas (TIG) equipment (Ref. 70-00-00) and a small diameter filler rod (PWC05-151) to minimize the weld bead size. Stress relief after welding is not necessary.

NOTE: Use a copper back-up plate to reduce distortion with the addition of argon gas backup, if available.

Table 201 Determination of Filler Material for Crack Repairs

Combustion Chamber Liner Part No.	Filler Metal	PT6A-21	PT6A-27	PT6A-28
3014144 *	AMS 5798		X	X
3020491 *	AMS 5798		X	X
3011283 *	AMS 5798		X	X
3022535	AMS 5837		X	X
3024576	AMS 5837	X	X	X
* Pre-SB1216 and Pre-SB1220				

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